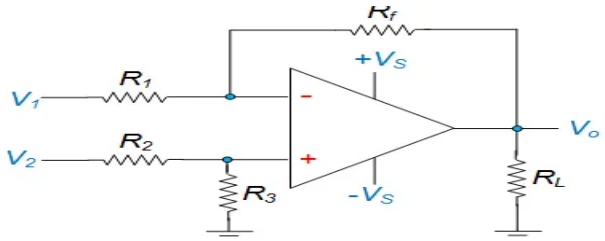
Chapter two

Differential amplifier

What is a Differential Amplifier?

The term differential comes from the amplifier's ability to amplify the difference of two input signals applied to its inputs. A differential amplifier (also known as a difference amplifier or op-amp subtractor) **is a type of electronic amplifier that amplifies the difference between two input voltages.** A differential amplifier is an analog circuit with two inputs (V_1 and V_2) and one output (V_0) in which the output is ideally proportional to the difference between the two voltages.



The formula for a simple differential amplifier can be expressed:

$$Vo = A_d (V1-V2)$$

Where

- V_0 is the output voltage
- V_1 and V_2 are the input voltages
- A_d is the gain of the amplifier (i.e. the differential amplifier gain)

From the formula above, you can see that when any difference between inputs V_1 and V_2 is multiplied (i.e. amplified) by the differential amplifier gain A_d. This is why the differential amplifier is also known as a difference amplifier – the difference between the input voltages is amplified.

There are two different types of differential amplifier circuits:

1. BJT Differential Amplifier – This is a differential amplifier built using transistors, either Bipolar Junction Transistors (BJTs) or Field Effect Transistors (FETs)
2. Opamp Differential amplifiers built using Operational Amplifiers

BJT Differential Amplifier

Figure 1 shows such a BJT differential amplifier circuit made of two BJTs (Q_1 and Q_2) and two power supplies of opposite polarity, V_{CC} and $-V_{EE}$ which uses three resistors among which two are the collector resistors, R_{C1} and R_{C2} (one for each transistor) while one is the emitter resistor R_E common to both transistors.

Here the input signals (V_1 and V_2) are applied to the base of the transistors while the output is collected across their collector terminals (V_{o1} and V_{o2}). The circuit diagram for a BJT differential amplifier is shown below:

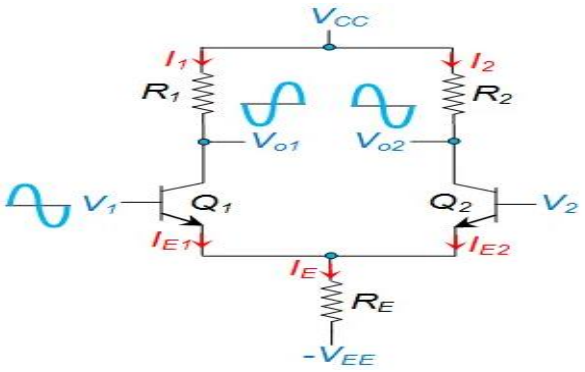


Figure 1. BJT differential amplifier

In this case, if the V_1 at Q_1 is sinusoidal, then as V_1 goes on increasing, the transistor starts to conduct and this results in a heavy collector current I_{C1} increasing the voltage drop across R_{C1}, causing a decrease in V_{o1}. Due to the same effect, even I_{E1} increases which increases the common emitter current, I_E resulting in an increase of voltage drop across R_E. This means that the emitters of both transistors are driven towards positive which in turn implies that the base of Q_2 would start to become more and more negative. These results in a decrease of collector current, I_{C2} which in turn decreases the volt age drop across the collector resistor R_{C2}, resulting in an increase in the output voltage V_{o2}.

This indicates that the changes in the sinusoidal signal observed at the input of transistor Q_1 are reflected as such across the collector terminal of Q_2 and appear with a phase difference of 180° across the collector terminal of Q_1. The differential amplification can be driven by considering the output in-between the collector terminals of the transistors, Q_1 and Q_2.

Op-amp Differential Amplifier

An Op-Amp operating in differential mode can readily act as a **subtractor amplifier** as it results in an output voltage given by:

$$V_0 = A_d(V_1 - V_2)$$

Where V_1 and V_2 represent the voltages applied at its inverting and non-inverting input terminals (can be taken in any order) and A_d refers to its differential gain. As per this equation, the output of the Op-amp must be zero when the voltages applied at its terminals are equal to each other.

However, practically it will not be so as the gain will not be the same for both of the inputs. Thus, in a practical scenario, the mathematical expression for the output of the subtractor amplifier can be given as:

$$V_0 = A_d(V_1 - V_2) + A_C\left(\frac{V_1 + V_2}{2}\right)$$

Where A_C is called the common-mode gain of the amplifier. Thus, functionally-good difference amplifiers are expected to exhibit a high common-mode rejection ratio (CMRR) and high impedance.

However, it is to be noted that an Op-Amp can be suitably configured to result in a much practical **differential amplifier**, as shown in Figure 2. If closely observed, one can note that this circuit is just a combination of inverting and non-inverting amplifiers.

Hence its output voltage will be equal to the sum of the output voltages produced by the Op-Amp circuit operating as an inverting amplifier and the Op-Amp circuit operating as a non-inverting amplifier. Thus, one gets:

$$V_0 = -\frac{R_f}{R_1}V_1 + V_2\frac{R_f}{R_2 + R_3}\left(1 + \frac{R_f}{R_1}\right)$$

Now, if $R_1 = R_2$ and $R_3 = R_f$, then:

$$V_0 = -\frac{R_f}{R_1}V_1 + V_2\frac{R_f}{R_1 + R_f}\left(\frac{R_1 + R_f}{R_1}\right)$$

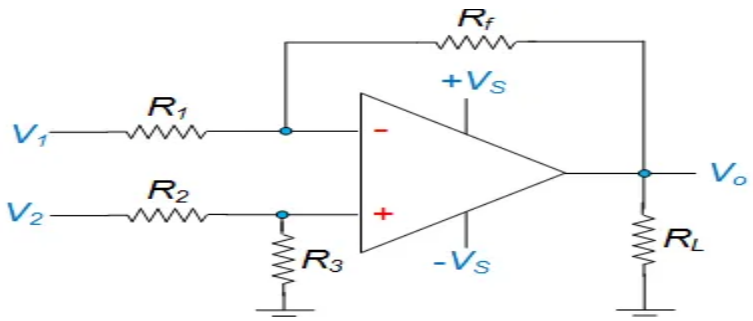


Figure 2. Differential amplifier using an op-amp

$$V_0 = -\frac{R_f}{R_1}V_1 + V_2\frac{R_f}{R_1}$$
$$V_0 = -\frac{R_f}{R_1}(V_1 - V_2)$$

This implies that the gain of the **differential amplifier** circuit shown in Figure 2 is given b

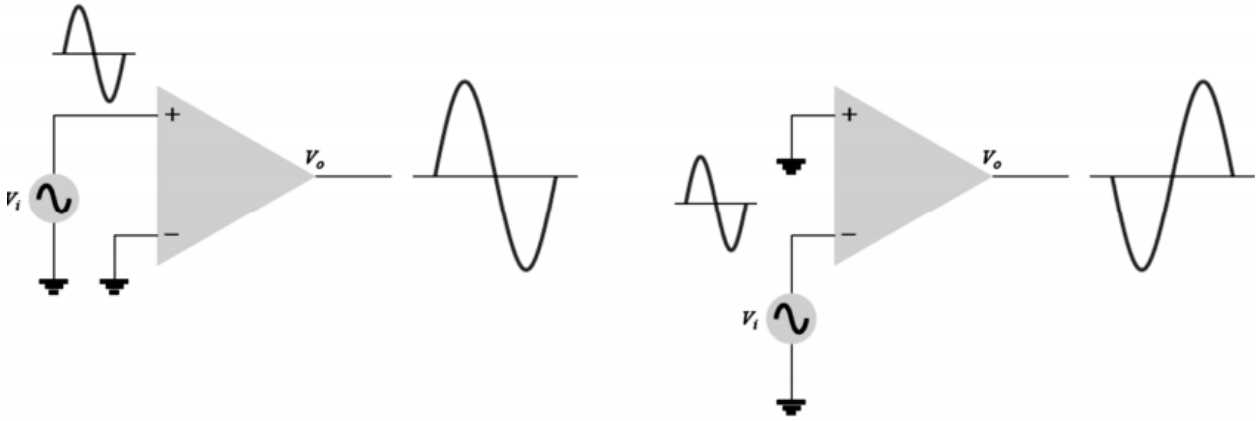
$$-\frac{R_f}{R_1}.$$

In addition, it is to be noted that the basic circuit shown in Figure 2 can be modified in many ways resulting in various circuit designs including the Wheatstone bridge differential amplifier, light-activated **subtractor amplifier,** and instrumentation amplifier. These devices are used as motor and/or servo controllers, signal amplifiers, analog multipliers, switches, volume controllers, automatic gain controllers, amplitude modulators, etc., and cover a wide range of applications including those in instrumentation systems, microphones, analog to digital converters and myriad applications.

Op-Amp Operation Modes:

The differential amplifier exhibits three modes of operation based on the type of input (and/or output) signals. These modes are ***single-ended***, ***double-ended*** or ***differential***, and ***common***. Since the differential amplifier is the input stage of the op-amp, the op-amp exhibits the same modes.

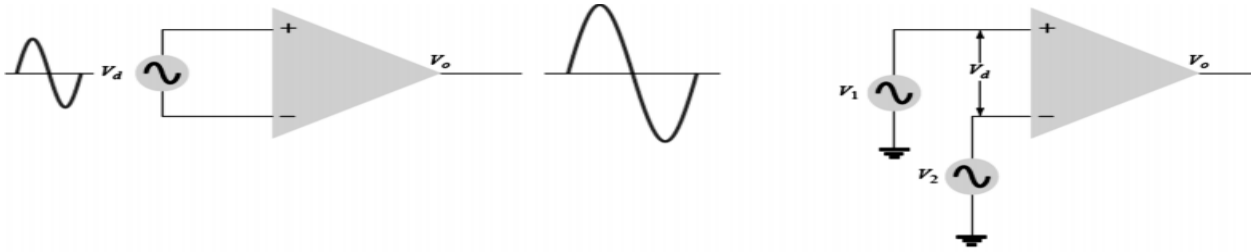
Single-Ended Input: Single-ended input operation results when the input signal is connected to one input with the other input connected to ground. Fig. 1-4 shows the signals connected for this operation. In Fig. 1-4(a), the input is applied to the plus input (with minus input at ground), which results in an output having the same polarity as the applied input signal. Fig. 1-4(b) shows an input signal applied to the minus input, the output then being opposite in phase to the applied signal.



(a) (b)

Fig. 1-4

Double-Ended (Differential) Input: In addition to using only one input, it is possible to apply signals at each input-this being a double-ended operation. Fig. 1-5(a) shows an input, V_d, applied between the two input terminals (recall that neither input is at ground), with the resulting amplified output in phase with that applied between the plus and minus inputs. Fig. 1.5(b) shows the same action resulting when two separate signals are applied to the inputs, the difference signal being $V_{i1} - V_{i2}$.



(a) (b)

Fig. 1-5

Double-Ended Output: While the operation discussed so far had a single output, the op-amp can also be operated with opposite outputs, as shown in Fig. 1-6(a). An input applied to either input will result in outputs from both output terminals, these outputs always being opposite in polarity. Fig. 1-6(b) shows a single-ended input with a double ended output. As shown, the signal applied to the plus input results in two amplified outputs of opposite polarity. Fig. 1-6(c) shows the same operation with a single output measured between output terminals (not with respect to ground). This difference output signal is $V_{o1} - V_{o2}$. The difference output is also referred to as a floating signal since neither output terminal is the ground (reference) terminal. Notice that the difference output is twice as large as either V_{o1} or V_{o2} since they are of opposite polarity and subtracting them results in twice their amplitude. Fig. 1-6(d) shows a differential input, differential output operation. The input is applied between the two input terminals and the output taken from between the two output terminals. This is fully differential operation.

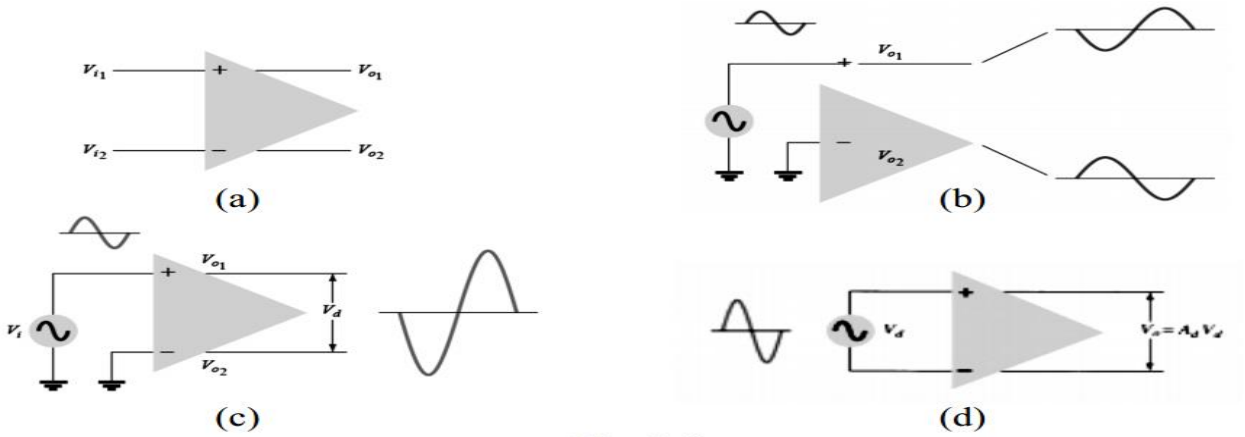


Fig. 1-6

Common-Mode Operation: When the same input signals are applied to both inputs, common-mode operation results, as shown in Fig. 1-7. Ideally, the two inputs are equally amplified, and since they result in opposite polarity signals at the output, these signals cancel, resulting in 0-V output. Practically, a small output signal will result

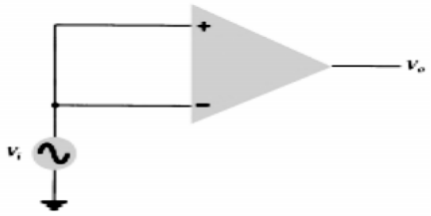


Fig. 1-7

Common-Mode Rejection: A significant feature of a differential connection is that the signals which are opposite at the inputs are highly amplified, while those which are common to the two inputs are only slightly amplified-the overall operation being to amplify the difference signal while rejecting the common signal at the two inputs. Since noise (any unwanted input signal) is generally common to both inputs, the differential connection tends to provide attenuation of this unwanted input while providing an amplified output of the difference signal applied to the inputs. This operating feature, referred to as common-mode rejection.

1.4 Common-Mode Reject Ratio (CMRR):

One of the more important features of a differential circuit connection, as provided in an op-amp, is the circuit's ability to greatly amplify signals that are opposite at the two inputs, while only slightly amplifying signals that are common to both inputs. An op-amp provides an output component that is due to the amplification of the difference of the signals applied to the plus and minus inputs and a component due to the signals common to both inputs. Since amplification of the opposite input signals is much greater than that of the common input signals, the circuit provides a common mode rejection as described by a numerical value called the common-mode rejection ratio (CMRR).

Differential Inputs: When separate inputs are applied to the op-amp, the resulting difference signal is the difference between the two inputs.

$$V_d = (V_{i1}-V_{i2})$$

Common Inputs: When both input signals are the same, a common signal element due to the two inputs can be defined as the average of the sum of the two signals.

$$Vc = (V_{i1}+V_{i2}) / 2$$

Output Voltage: Since any signals applied to an op-amp in general have both in-phase and out-of phase components, the resulting output can be expressed as

$$Vo = AdVd+AcVc$$

where A_d = differential gain, and A_c = common-mode gain of the amplifier.

Having obtained A_d and A_c, we can now calculate a value for the common-mode rejection ratio (CMRR), which is defined by the following equation:

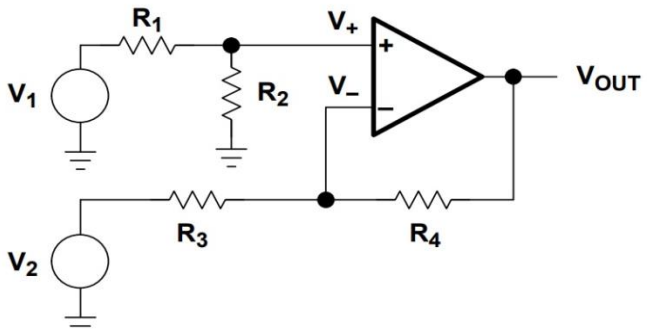
$$CMRR = A_d/A_c$$

The value of CMRR can also be expressed in logarithmic terms as

$$CMRR\ (dB) = 20*Log\ (A_d/A_c)$$

Practical Differential Amplifiers

The following image shows a simple Differential Amplifier using an Op Amp. Here, V_1 is the Non-Inverting Input Voltage, V_2 is the Inverting Input Voltage and V_{OUT} is the Output Voltage.



If you observe the above circuit of the difference amplifier, it is a combination of both the Inverting Amplifier and the Non-Inverting Amplifier. So, to calculate the output voltage of a Differential Amplifier, we will use both the Inverting and Non-Inverting outputs and add them together.

Calculating the Output Voltage (Detail note included under chapter three**)**

Let V_+ be the voltage at the Non-Inverting terminal and V_- be the voltage at the Inverting Terminal of the above Differential Amplifier Circuit. We can calculate the value of V_+ using the Potential Divider Rule.

Resistors R_1 and R_2 form a Voltage Divider Network with V_1 as the Input Voltage and V_+ as the output voltage and this V_+ is applied at the non-inverting terminal. So,

$$V_+ = V_1 (R_2 / R_1 + R_2)$$

If V_+ is the input to the non-inverting terminal and G_+ is the gain of the Non-Inverting Amplifier, then non-inverting output V_{OUT+} is given by:

$$V_{OUT+} = V_+ G_+$$

From the above circuit, we can calculate the Non-Inverting Gain G_+ as:

$$G_+ = (R_3 + R_4) / R_3 = 1 + (R_4 / R_3)$$

Using the values of V_+ and G_+ in the equation of V_{OUT+}, we get

$$V_{OUT+} = V_1 (R_2 / R_1 + R_2) (1 + (R_4 / R_3))$$

Coming to the Inverting Output V_{OUT-}, we have to calculate it with respect to the inverting input V_2 and the Inverting Gain G_-.

$$V_{OUT-} = V_2 G_-$$

From the above circuit, we can calculate the Inverting Gain G_- as:

$$G_- = - R_4 / R_3$$

So, V_{OUT-} is given by:

$$V_{OUT-} = V_2 (- R_4 / R_3)$$

We have both V_{OUT+} and V_{OUT-} values. To get the final V_{OUT} value, we have to add these values.

$$V_{OUT} = V_{OUT+} + V_{OUT-}$$

$$V_{OUT} = V_1 (R_2 / R_1 + R_2) (1 + (R_4 / R_3)) – V_2 (R_4 / R_3)$$

This is the output voltage of a Differential Amplifier. The above equation looks complex. So, to reduce the complexity and simply the equation, let us take a special case where $R_3 = R_1$ and $R_4 = R_2$.

If we apply these values in the above equation, we the output voltage as:

$$V_{OUT} = R_2 / R_1 (V_1 – V_2) = R_4 / R_3 (V_1 – V_2)$$

Now, from this equation, it is clear that the differential voltage $(V_1 – V_2)$ is multiplied by the gain R_2 / R_1. Hence, it is Differential Amplifier.

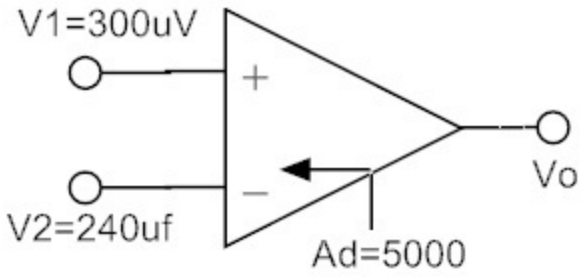
Example

Determine the output voltage of a differential amplifier for the input voltages of 300 µV and 240 µV. The differential gain of the amplifier is 5000 and the value of CMRR is

(i) 100

(ii) 10^5

The differential amplifier for the given data is represented as shown in the figure.



For CMRR = 100:

$$CMRR = A_D / A_C$$

$$100 = 5000 / A_C$$

$$So, A_C = 50$$

Difference Mode Voltage V_{DM}

$$V_{DM} = V_1 - V_2 = 300\ \mu V - 240\ \mu V = 60\ \mu V$$

Common Mode Voltage V_{CM}

$$V_{CM} = (V_1 + V_2) / 2 = 540\ \mu V / 2 = 270\ \mu V$$

Output Voltage V_{OUT}

$$V_{OUT} = A_D V_{DM} + A_C V_{CM}$$

$$= 5000 \times 60\ \mu V + 50 \times 270\ \mu V$$

$$V_{OUT} = 313500\ \mu V = 313.500\ mV$$

For CMRR = 10^5:

$$A_C = A_D / CMRR = 5000 / 10^5 = 0.05$$

$$V_{OUT} = A_D V_{DM} + A_C V_{CM} = 5000 \times 60\ \mu V + 0.05 \times 270\ \mu V$$

$$V_{OUT} = 300013.5\ \mu V = 300.0135\ mV$$

Exercise

Calculate the CMRR and express it in decibel for the circuit measurements shown in Fig. 1-8. [Answers: 666.7, 56.48 dB]

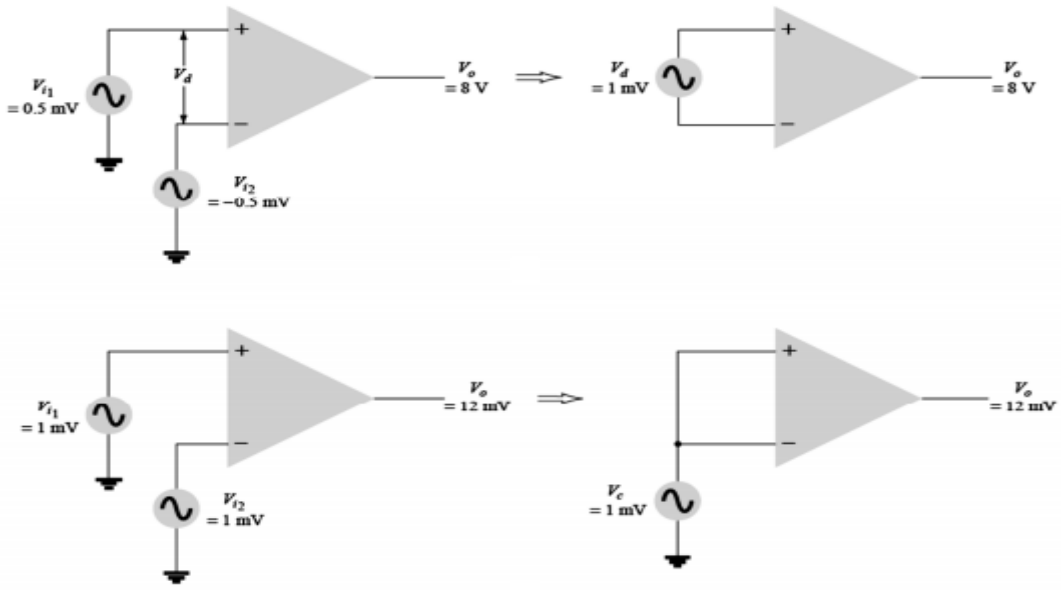


Fig. 1-8